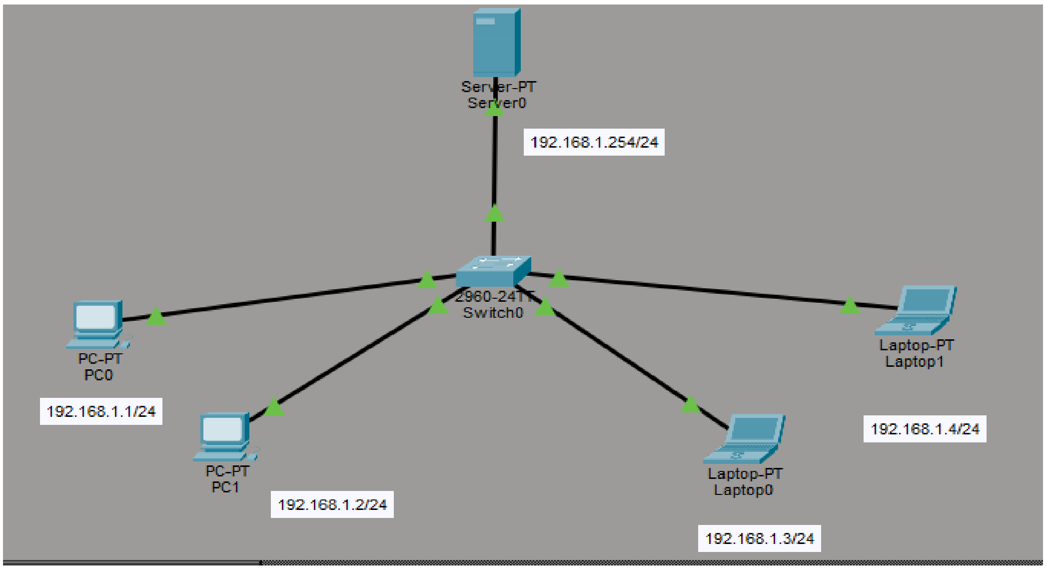
Lab-2

1. To perform a DNS lab using Cisco Packet Tracer



Network Components

1. DNS Server (Server0)
IP Address: 192.168.1.254/24
Function:
Stores domain name records.
Converts domain names (e.g., www.anudeep.com) into IP addresses.
Responds to DNS queries from client devices.

2. Switch (2960-24TT)
Function:
Connects all devices in the Local Area Network (LAN).
Forwards data packets between devices based on MAC addresses.
Acts as a central communication device.

3. Client Devices

Device	IP Address
PC0	192.168.1.1/24
PC1	192.168.1.2/24
Laptop0	192.168.1.3/24
Laptop1	192.168.1.4/24

All clients are configured in the same subnet (255.255.255.0).

How DNS Works in This Network
Suppose PC0 wants to access a website named:
www.anudeep.com

Step 1: User enters domain name
PC0 sends a DNS request asking:
"What is the IP address of www. anudeep.com?"

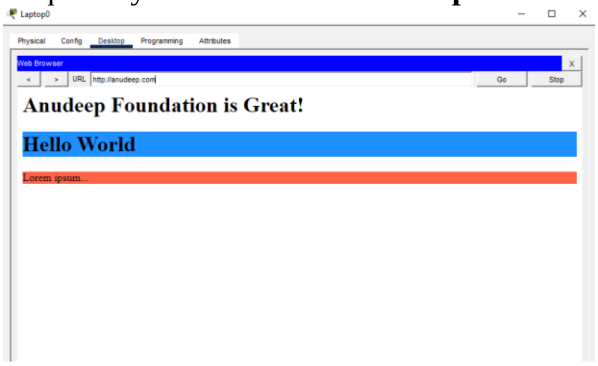
Step 2: Request reaches DNS Server
The request travels:
PC0 ➔ Switch ➔ DNS Server

Step 3: DNS Server checks records
The DNS server looks into its DNS database.

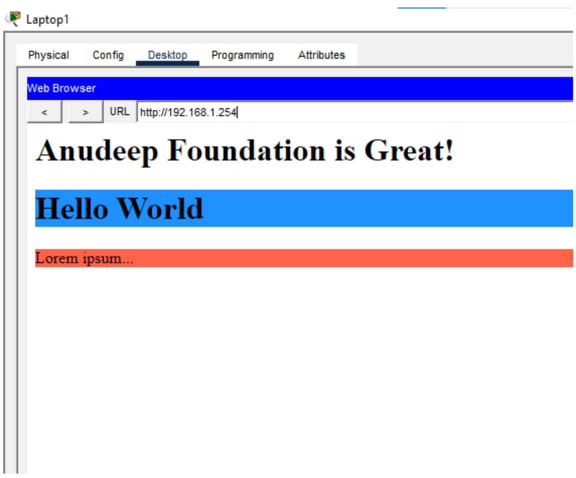
Step 4: DNS Reply
The server sends back the corresponding IP address.

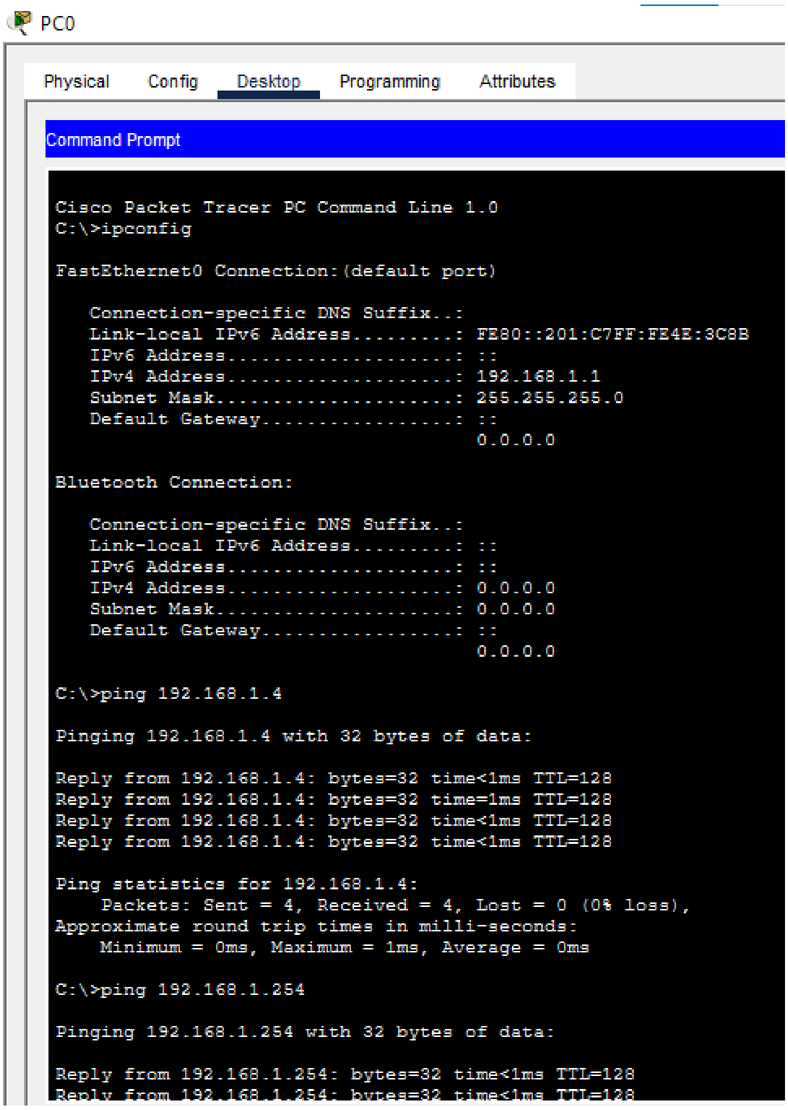
Step 5: Communication Begins
Now PC0 knows the destination IP address and can communicate with that device directly.

Request by Server Name : **Anudeep.com**



Request by Server IP address : **192.168.1.254**






```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:C7FF:FE4E:3C8B
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.1
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time<1ms TTL=128
Reply from 192.168.1.254: bytes=32 time<1ms TTL=128
```